

# Evaluating RAG Beyond Retrieval Accuracy

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## ABSTRACT

Retrieval-augmented generation connects language generation to external evidence [1], but a relevant passage does not guarantee a correct answer. This technical study proposes an evaluation protocol that separates retrieval, generation, and operational behavior. Its contribution is an experimental design, not a new algorithm or a completed benchmark. No measured results are claimed.

**Index terms:** retrieval-augmented generation, evaluation, grounded generation, reproducibility.

## I. PROBLEM FORMULATION

Consider an assistant answering questions over a versioned document collection. Failures can arise because the retriever misses evidence, the generator misuses available evidence, or the application retrieves material outside the user's access scope. A single aggregate answer score obscures these different causes.

Ragas describes reference-free evaluation across context relevance, faithfulness, and answer quality [2]. Such automated judgments are useful for iteration, but this proposed protocol also includes a human-reviewed holdout set. A model judge can share the biases and blind spots of the model it evaluates.

## II. PROPOSED EXPERIMENTAL DESIGN

Freeze the corpus snapshot, parser, chunking policy, embedding revision, index settings, prompt, and generation model. Record a checksum for the corpus and a configuration identifier for every run. Split by source document or document family to reduce near-duplicate leakage between development and evaluation.

Construct questions in four groups: directly answerable, requiring multiple passages, unanswerable from the corpus, and adversarial or access-restricted. Store expected evidence identifiers and an explicit answerability label. Keep the final holdout inaccessible during prompt tuning.

Compare a lexical baseline, dense retrieval, and a hybrid configuration. Change one component at a time; otherwise a better answer cannot be attributed to a better retriever. Evaluate identical queries and report paired differences with uncertainty intervals rather than just a leaderboard.

## III. MEASUREMENT CONTRACT

| Layer      | Measure                                           | Interpretation                                          |
|------------|---------------------------------------------------|---------------------------------------------------------|
| Retrieval  | Recall@k                                          | Fraction of annotated relevant units recovered          |
| Ranking    | nDCG@k                                            | Whether highly relevant evidence appears early          |
| Generation | Supported-claim fraction                          | Claims supported by retrieved evidence                  |
| Citations  | Citation precision                                | Cited passages actually supporting the associated claim |
| Abstention | False-answer rate on unanswerable questions       | Unsupported answers when evidence is absent             |
| Operations | p50/p95 latency, errors, cost per successful task | User-visible reliability and resource use               |

Define the evidence unit before scoring. Passage-level and document-level recall answer different questions. Review a sample of citation judgments manually and record disagreement. A claim without a citation should not silently disappear from the denominator.

#### IV. RELEASE DECISION

Pre-register acceptance criteria against the intended use case. A candidate should pass the access-isolation tests and avoid materially worsening unsupported-answer rates before speed or cost improvements count as a release benefit. Report cold and warm runs separately, and retain failed requests in reliability reporting.

For an illustrative test plan, a team might allocate 200 held-out queries across the four groups and repeat each configuration with three seeds. These are proposed design parameters, not an executed experiment or a statistically guaranteed sample size. Power and coverage depend on expected effect size and the risk of missed failures.

#### V. LIMITATIONS AND NEXT STEPS

This protocol cannot establish performance until implemented on a representative corpus. Labels may be incomplete, automated graders imperfect, and production traffic different from a fixed test set. Next steps are to publish an appropriately licensed corpus manifest, annotation guide, run configurations, and per-query outputs; then reproduce the comparison with uncertainty estimates.

**Publication status:** Independent technical study, prepared for this portfolio with AI assistance. Not peer-reviewed, IEEE-published, or an IEEE compliance certification. Numbered references use an IEEE-style presentation.

#### REFERENCES

- [1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," 2020. arXiv:2005.11401.
- [2] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "Ragas: Automated Evaluation of Retrieval Augmented Generation," 2023, revised 2025. arXiv:2309.15217.